

Personalized Educational Maze Game – A Case Study

Valentina Terzieva



**INSTITUTE OF INFORMATION AND COMMUNICATION TECHNOLOGIES
BULGARIAN ACADEMY OF SCIENCE**

- 1. Introduction**
- 2. Use of educational computer games in schools**
- 3. Educational maze games**
- 4. Educational enriched maze video game “Asenevtsi”**
- 5. Evaluation of an educational video maze game “Asenevtsi”**
- 6. Conclusion**

ICT (r)evolution

- Transformed manners of living, working, communicating, learning
- Impact on education:
 - ways of knowledge delivery and presentation
 - ways of information obtaining and knowledge acquisition,
 - technology-based pedagogical approaches
 - learners-centered teaching process
 - technology-based personalization
- Growing potential of educational games, gamification, game-based learning:
 - increased effectiveness
 - improved motivation and learning results
 - appealing with inherent imagery, colors, and dynamics
- Teachers have recognized the need for games purposefully intended for education

2. Use of educational computer games in schools

Essential features and advantages

- Provide learners with immediate feedback – supports and stimulus for learning;
- Motivate students (even in case of failure) and raise their self-esteem;
- Engage students by interactivity and provoke a curiosity;
- Enable students to do experiments and test their knowledge;
- Provide opportunities for teaching and real-time evaluation;
- Enable difficulty adaptation of playing tasks according to the learners' performance;
- Facilitate the learning process, engage attention and reduce frustration of students;
- Contribute to more efficient use of time compared to traditional lessons;
- Allow for development of soft skills (collaboration, teamwork, leadership, etc.).

2. Use of educational computer games in schools

Barriers and difficulties

- Lack of technological devices and time
 - Lack of appropriate educational games
 - Deficient IT competence of teachers
 - Few tools with limited functionality for construction educational games
 - No opportunity for adaption of available games
- ❖ Need for tools to facilitate the design and personalization of educational video games

Application aspects

- Educational games have proved as an efficient approach
- Suitable for most age groups, complementary, and even for basic training
- Integration of interactive educational games in traditional lessons and self-training

Educational video game of a type enriched maze

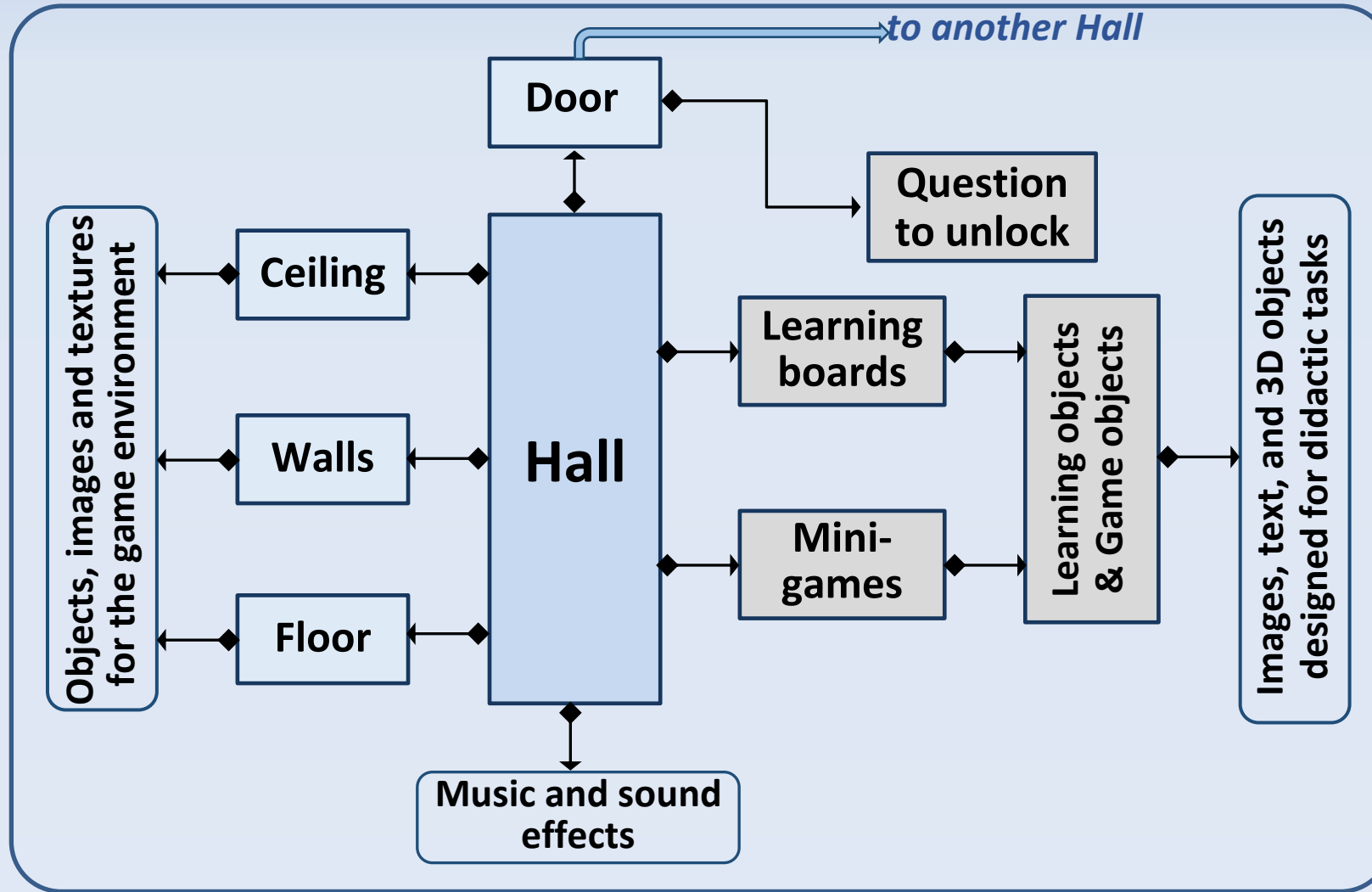
- 3D game consisting of several interconnected halls
- Built-in didactic mini-games (puzzles) and multimedia learning resources
- Visual environment for presenting learning content, information, and related resources
- Intended to achieve particular learning goals

Application aspects

- Practical tools for training purposes
- Suitable for implementation in different educational contexts
- Allow easy structuring and formal description of learning content
- Allow multiple learning paths – enable personalization and adaptation
- Allow integration of various kinds of didactic tasks embedded in mini-games
- Enable tailoring of didactic tasks according to the learner's characteristics

3. Educational maze games

Metamodel of an enriched maze educational video game



Educational video game of type enriched maze

Mini-games – fun-to-play, relatively simple, with strict rules

Puzzle-type mini-games – easily associated with different didactic tasks

- a particular mental challenge that have to be solved using knowledge, skills, logic, and ingenuity
- involve one player, has clear and relatively simple rules,
- the objective is to find a solution to well-defined logical or other tasks.

Combinations of puzzle-type mini-games in a maze game

- represent a variety of learning tasks
- help learners develop critical and logical thinking
- make the learning process interactive and more engaging

Pedagogical viewpoint on mini-games

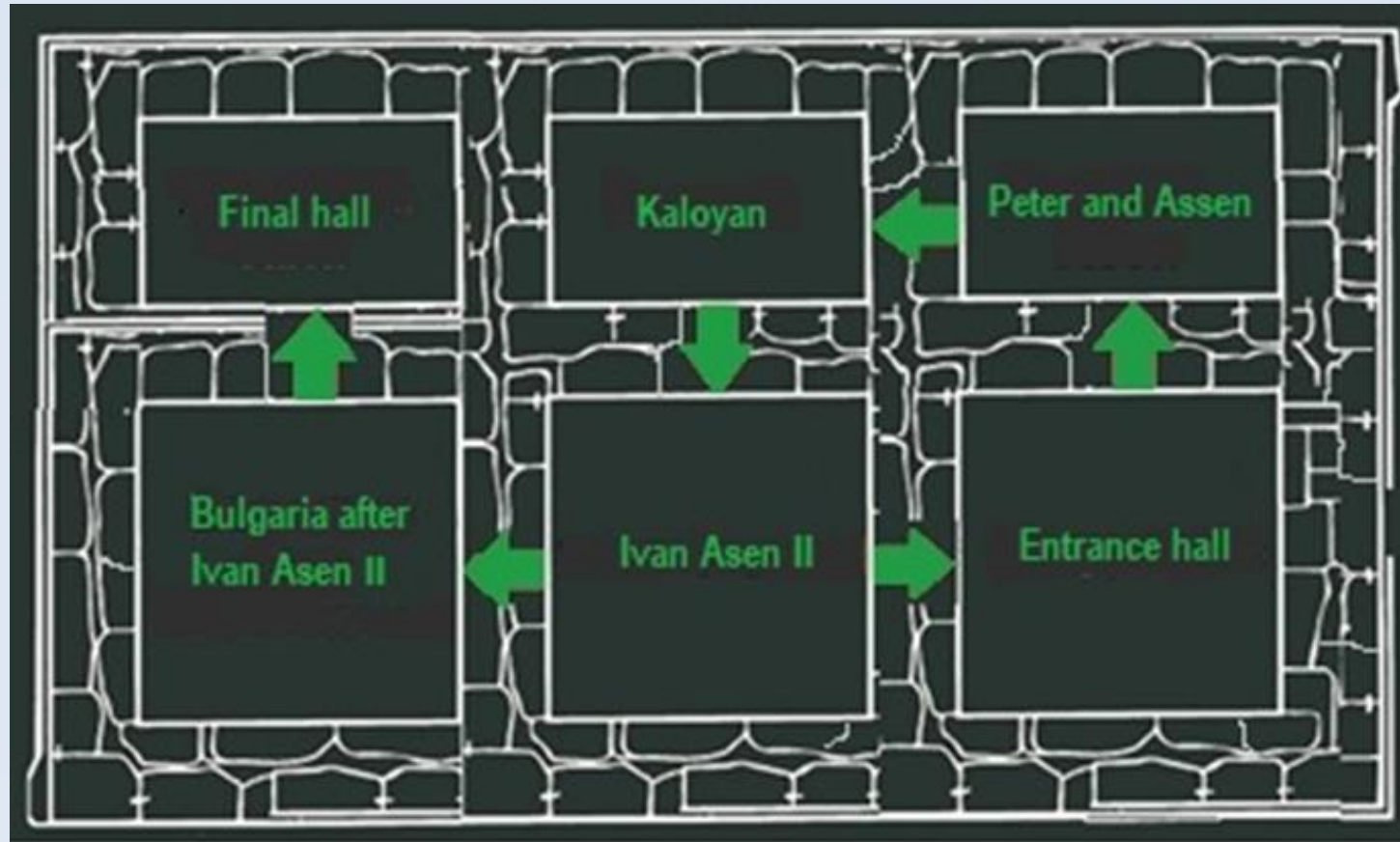
- Alternative form for delivering educational content
- Motivating and challenging learners
- Stimulating new activities
- Easy to integrate into learning practice
- Possess natural interactivity and adaptivity
- Suit various learning scenarios

Personalization of educational 2D and 3D mini-games (puzzles)

- **Answering questions:** *Personalization* – the type and difficulty level of the questions, the number of possible answers, and the hints provided;
- **Finding words** in a table of letters (word soup) – words can be placed in different directions and in straight and reverse order: *Personalization* – table size, the type and complexity of the words, and the availability and type of the hint provided;
- **Matching (memory) game** matching two by two tiles with images, text, etc.: *Personalization* – the number and content of tiles, matching criteria, and the availability and type of the hint provided;
- **Roll balls** with text and pictures to a corresponding position on a map on the floor or to a ring: *Personalization* – the images, number, and number and location of the places of balls;
- **Classification or arrangement** of several educational objects according to a particular characteristic: *Personalization* – the size, type, number, and grouping criterion;
- **Finding visible translucent objects** related to the game context: *Personalization* – the size, type, number, and location of the hidden objects.

4. Educational enriched maze video game “Asenevtsi”

Structure model of an educational video maze game “Asenevtsi”



4. Educational enriched maze video game “Asenevtsi”

Screenshots of universal and personalized game versions

“Find the Words”



“Memory Game”



4. Educational enriched maze video game “Asenevtsi”

Screenshots of universal and personalized game versions

“Rolling balls”



“Quiz”



4. Educational enriched maze video game “Asenevtsi”

Screenshots of mini-games

Finding hidden objects



Arranging objects



5. Evaluation of an educational video maze game “Asenevtsi”

Experimental settings

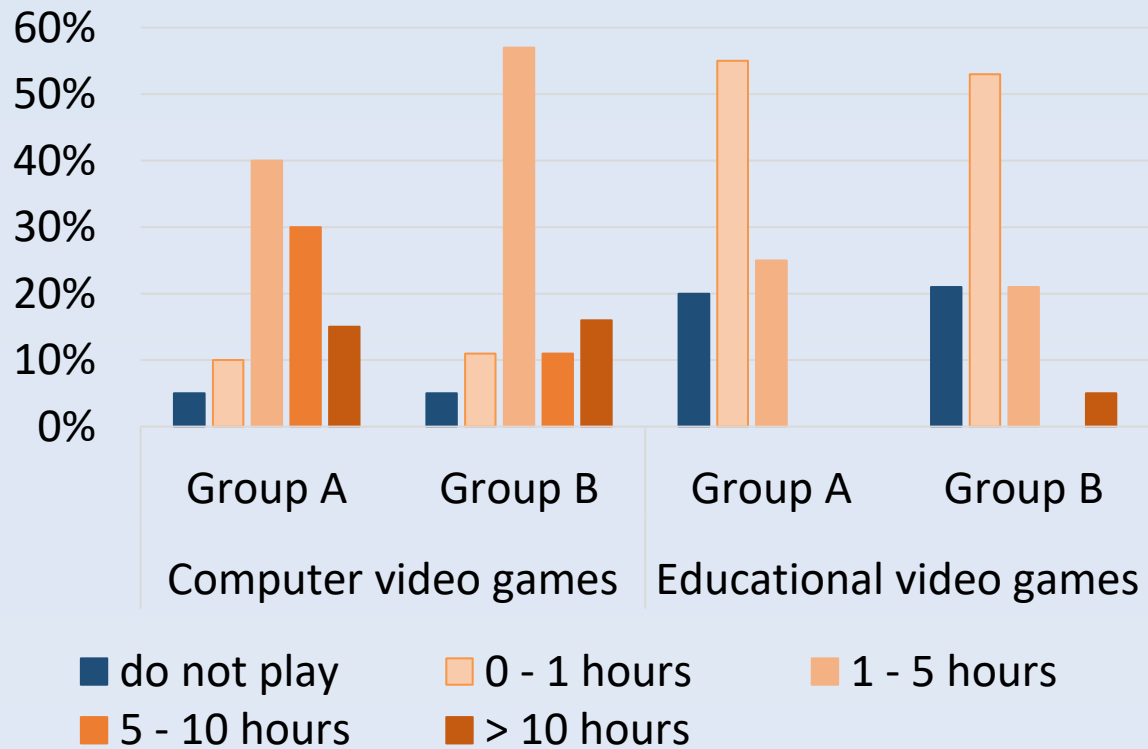
- 1. Selection of target groups of learners** – sixth-grade students
 - Group A: students with advanced knowledge level and deep interest in the game topic
 - Group B: students with a basic knowledge level who are not interested in the topic
- 2. Introducing the game to students** – presentation on how to play the game, types of mini-games, and the game’s goal
- 3. Experimental testing** – online game on a computer
 - Group A students play personalized game
 - Group B: students play universal game
- 4. Online survey** – students’ perception of the playing process: profiling questions, achieved results, gaming experience and game learnability
- 5. Processing and analysis of survey results** – evaluation of qualitative and quantitative characteristics

5. Evaluation of an educational video maze game “Asenevtsi”

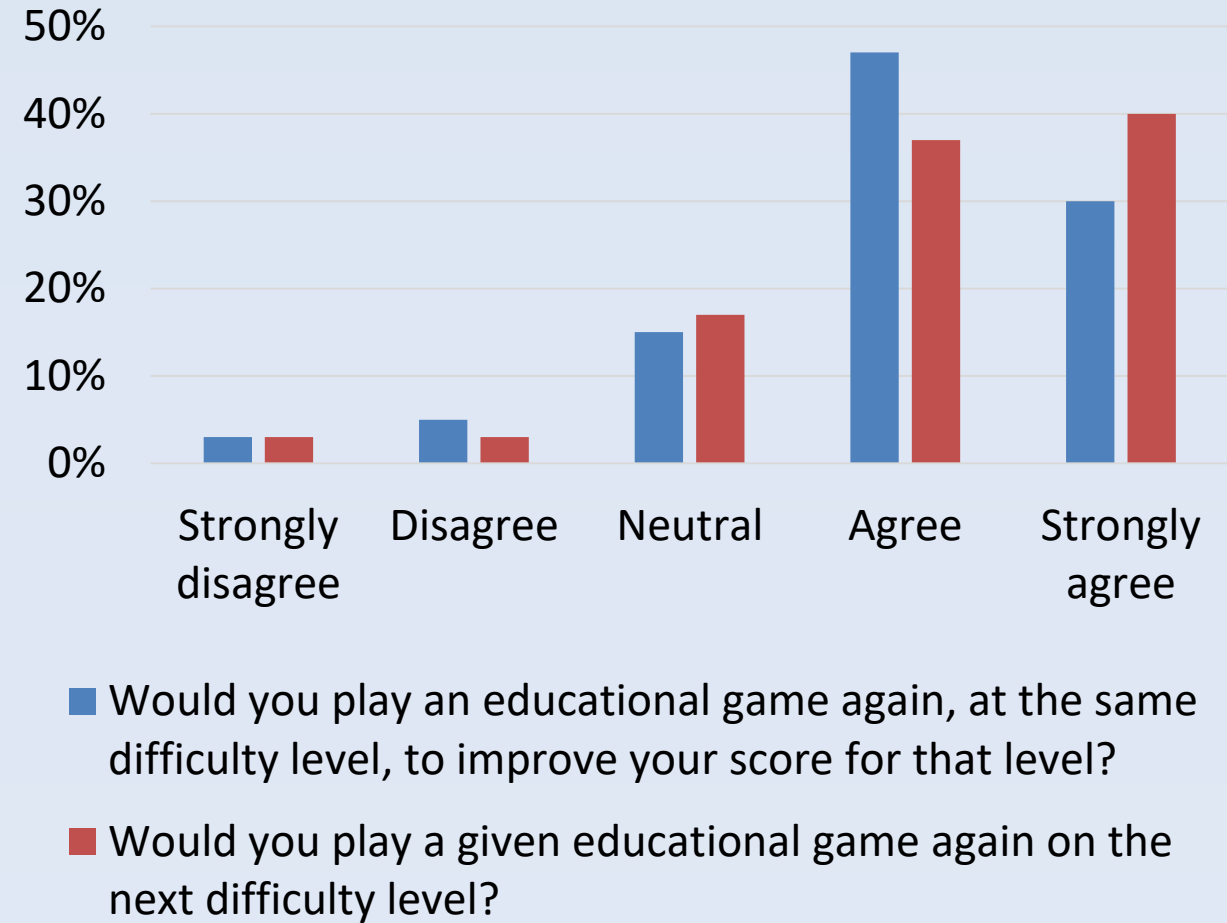
Profile of the participants

40 students: **Boys: 60%** **Girls: 40%**

Gaming experience



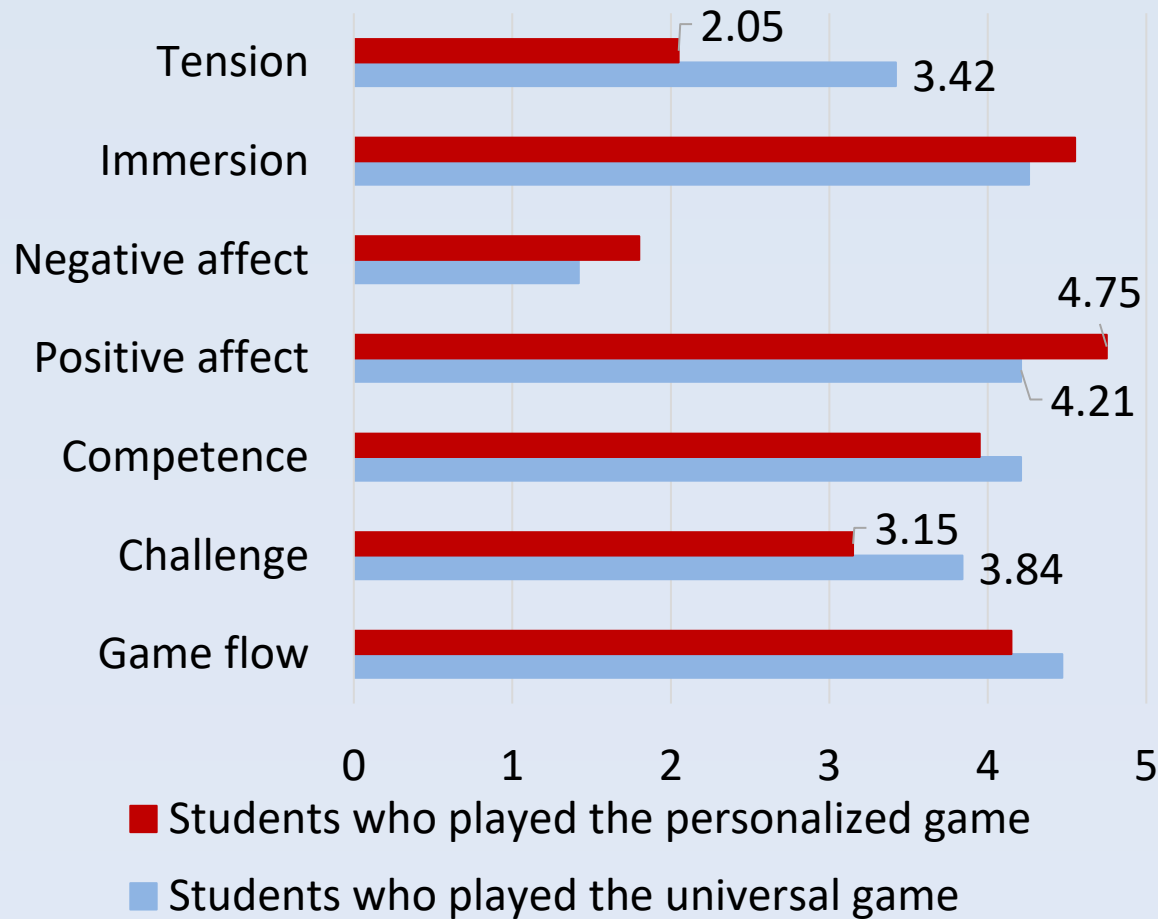
Attitude to GBL



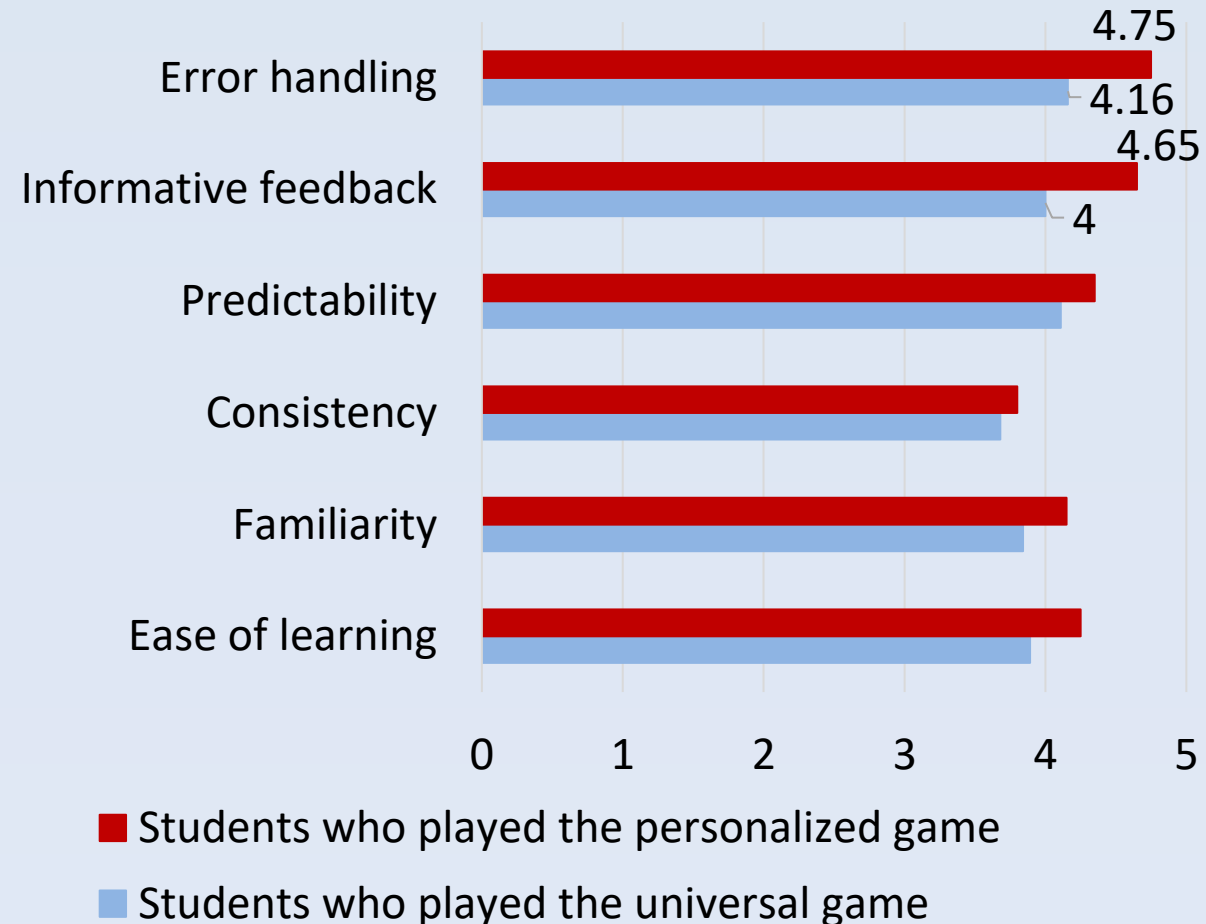
5. Evaluation of an educational video maze game “Asenevtsi”

Students’ assessment of the educational game “Asenevtsi”

Game experience



Learnability



5. Evaluation of an educational video maze game “Asenevtsi”

Overall score results achieved by students in the gameplay session

Criterion	Group A Students – played personalized game	Group B Students – played universal game
Average game playing time	36 min	34 min
Average score (% of max. points)	72%	77%
Number of students achieved over 70% score	50%	60%

Educational video game “Asenevtsi”

- Efficient learning tool that is rated highly in terms of suitability for learning and game experience.
- Personalized game provides more opportunities and motivation for learning than the universal.
- Both game versions are suitable for the corresponding group of students at which are intended.
- If an educational game is appropriate for the knowledge level, needs, and preferences of a particular group of students, it brings learning benefits to them.

Future work

- Experimental testing with larger groups of students
- Detailed statistical analysis of the game impact on different groups of students
- Investigation of the factors influencing the efficacy of both versions of the game
- Upgrade of the personalization methodology with approaches allowing adaptive customization

Thank you for your attention!

Acknowledgements

This research is funded by the project MOTIVATE (MOdeling inTelligent Instruments for adaptIVe and personAlized educaTional STEM video gamEs), funded by the Bulgarian National Science Fund, Grant Agreement No. КП-06-H85/2 dated 05/12/2024



Valentina Terzieva
valentina.terzieva@iict.bas.bg
IICT-BAS